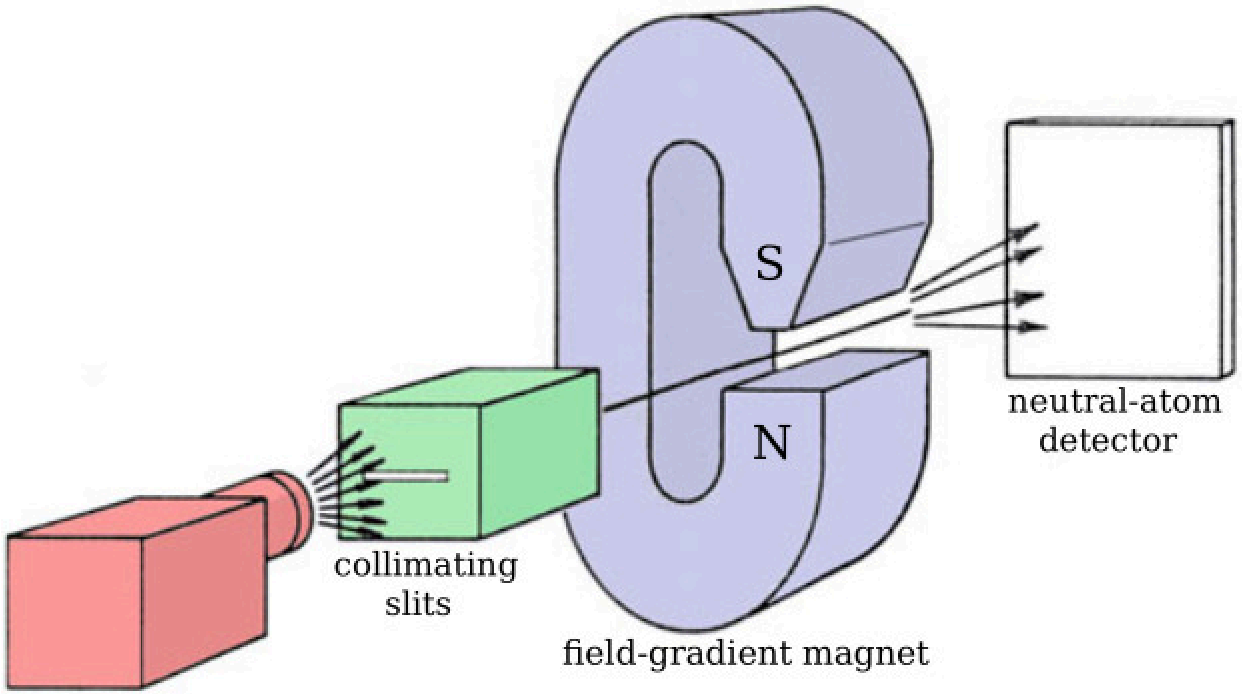
Stern-Gerlach Experiment

Luca Cicu - Lorenzo Liuzzo

27/11/2025

Experiment description

Apparatus


S
N
neutral-atom
detector
collimating
slits
field-gradient magnet
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Angular & Magnetic momenta

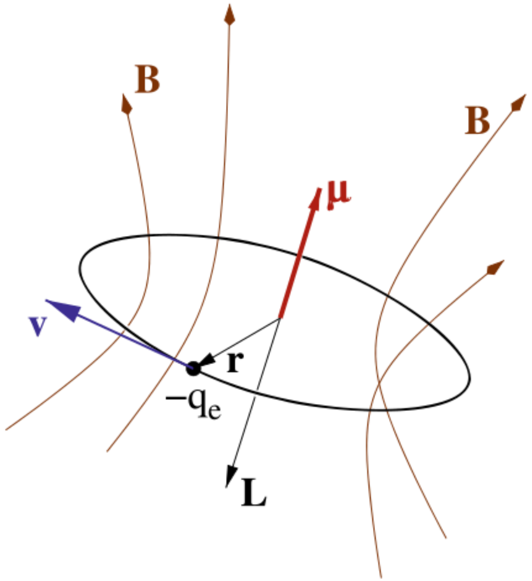

Figure 2: **Angular** and **magnetic** momenta generated by an electron orbiting circularly.

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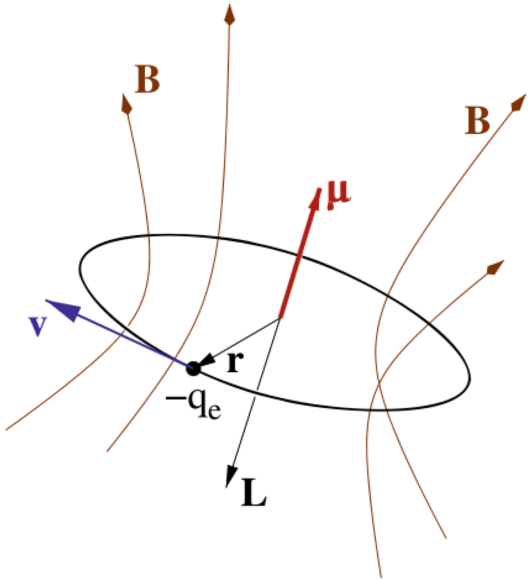

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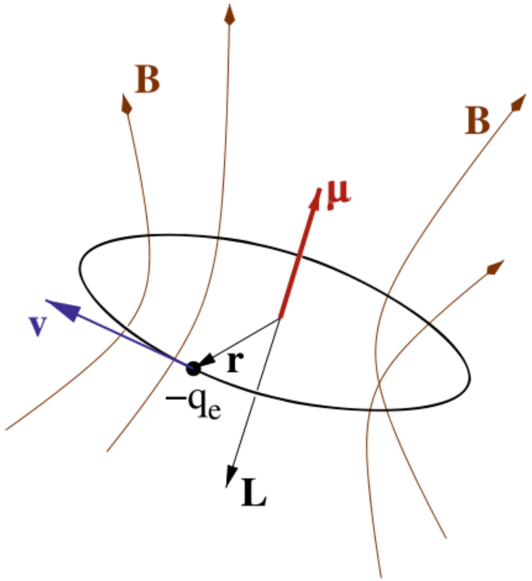

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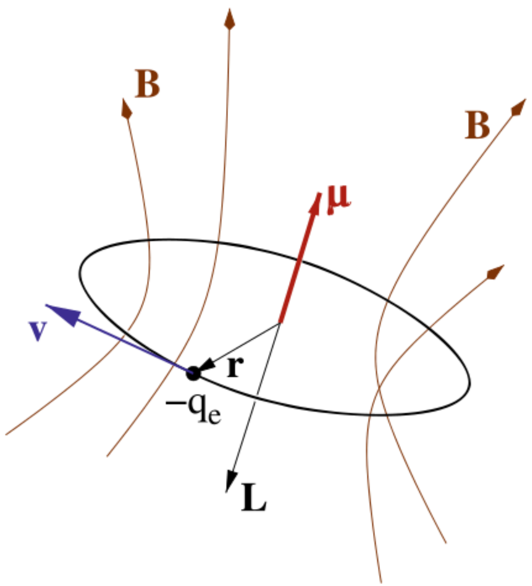

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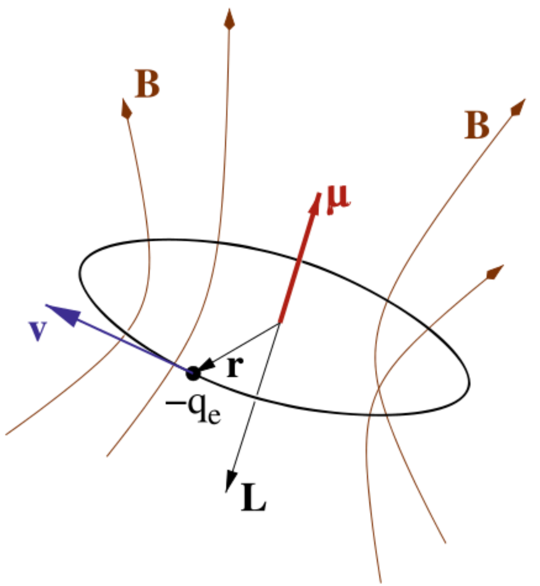

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Magnetic Hamiltonian

$$H_{\text{magn}} = -\boldsymbol{\mu} \cdot \boldsymbol{B}$$

Magnetic Interaction

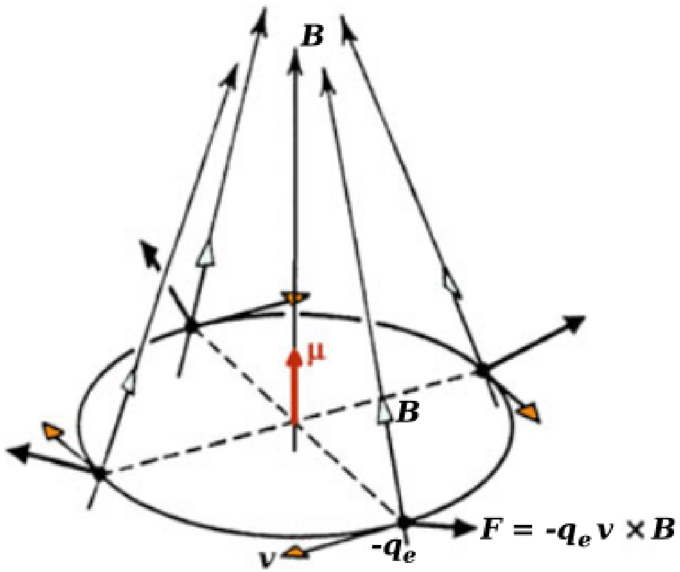

Figure 3: Microscopic origin of the magnetic force.

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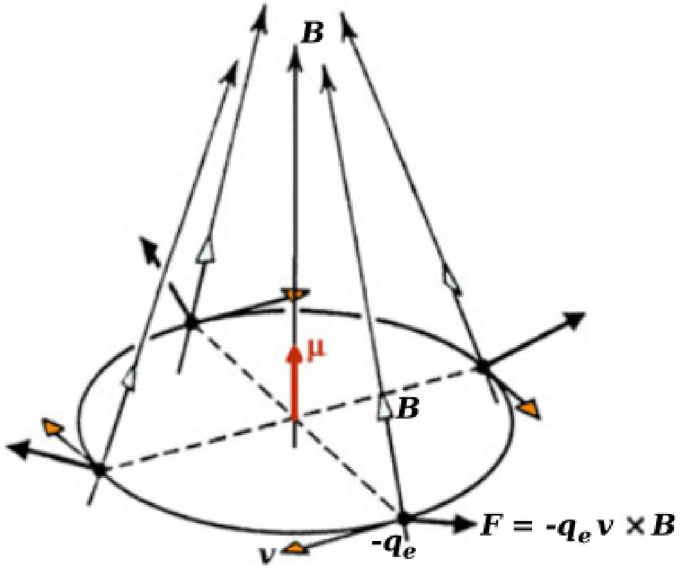

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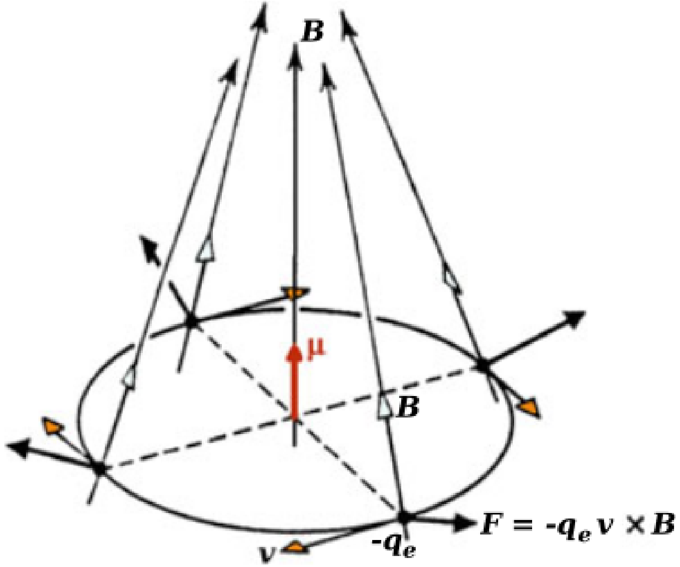

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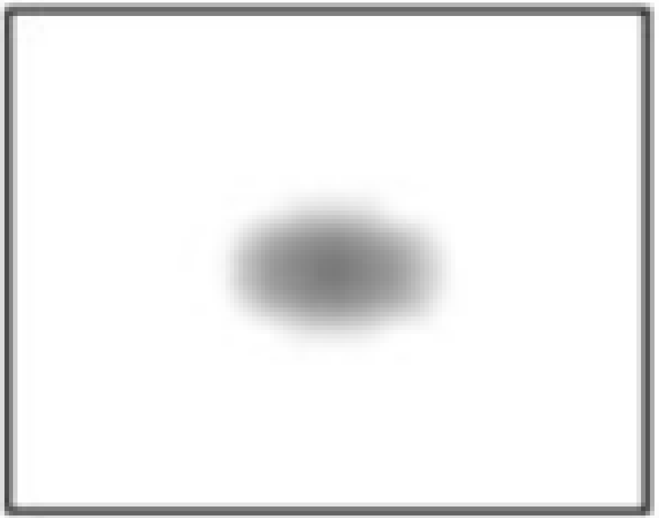
$$F_z = \mu_z \frac{\partial B_z}{\partial z} = m\ddot{z}$$

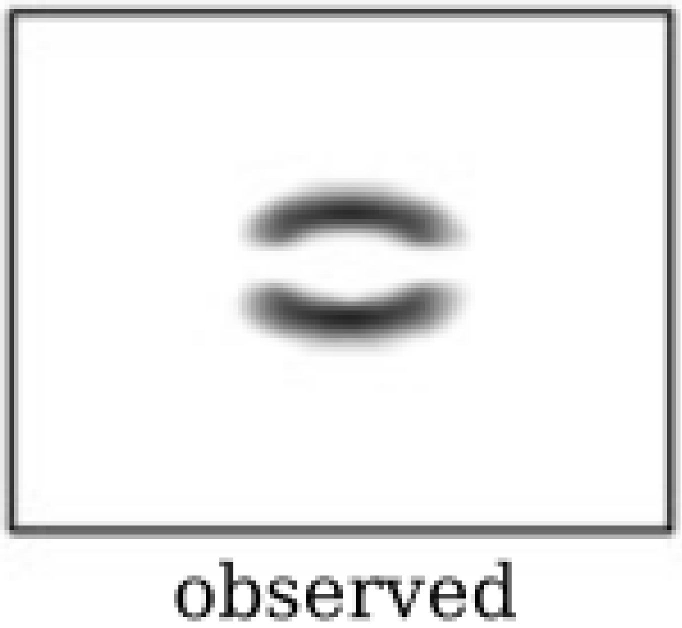
$$z(t) = z_0 + \dot{z}_0 t + \frac{\mu_z}{2m} \frac{\partial B_z}{\partial z} t^2$$

Classical Prediction



classically predicted

Observed Results



Why two deflected spots?

Historical Context

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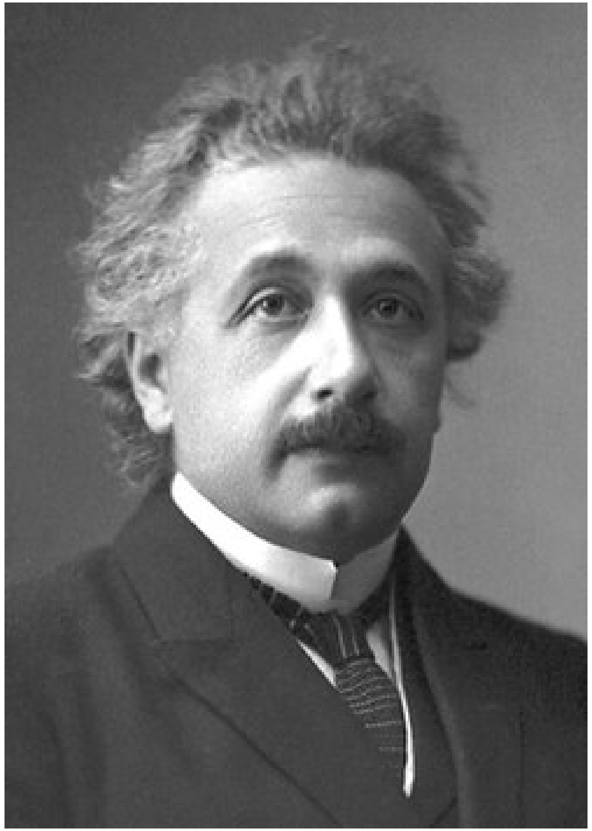

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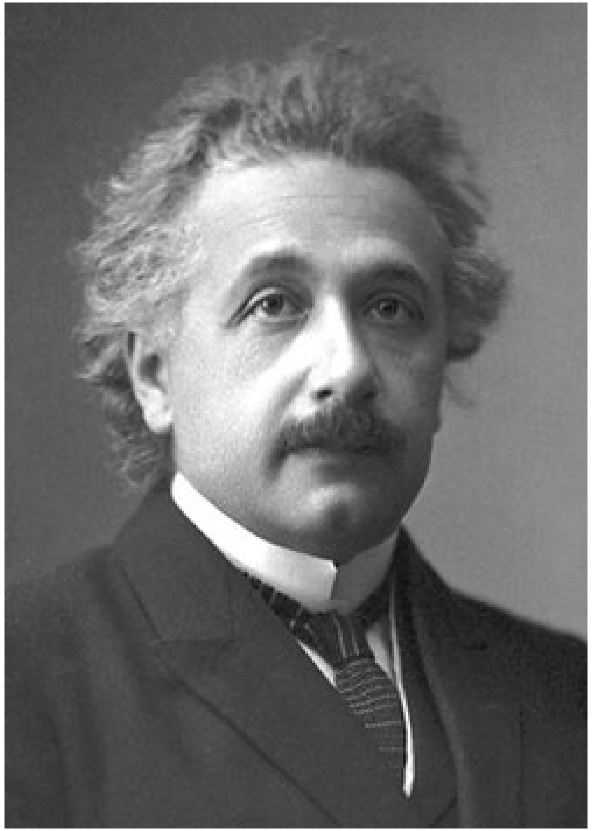

Quantum Mechanics

QM — States in Hilbert Space

$$|\psi\rangle \in \mathbb{H}$$

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- **Born's rule**

$$\langle\psi|\psi\rangle = \sum_i \left(c_i^\psi\right)^* c_i^\psi = \sum_i |c_i^\psi|^2 = 1$$

$|c_i^\psi|^2$ is the **probability** to find $|\psi\rangle$ in $|e_i\rangle$

QM — Operators

A **linear operator** $\hat{O}$ is a transformation from an Hilbert space into itself, that acts like

$$\hat{O}(\alpha|\psi\rangle + \beta|\varphi\rangle) = \alpha\hat{O}|\psi\rangle + \beta\hat{O}|\varphi\rangle = \alpha|\psi'\rangle + \beta|\varphi'\rangle$$

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We can also define the **adjoint** operator $\hat{O}^\dagger$: $\quad \langle\psi'| = \langle\psi|\hat{O}^\dagger \quad \hat{O}^\dagger_{ij} = \left(\hat{O}_{ji}\right)^*$

QM — Operators & Observables

We can associate every **observable** of a system to an operator

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Such operators are called **hermitian** and satisfy $\quad \hat{O}^\dagger_{ij} = \lambda_i^* \delta_{ij} = \lambda_i \delta_{ij} = \hat{O}_{ij}$

QM — Commutator of Operators

We define the **commutator** between two operators $\hat{A}$ and $\hat{B}$ as

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Two observables can be **simultaneously** determined iif their operators are compatible, that is if they share a **common basis** of eigenstates:

$$\hat{A}|e_i\rangle = \lambda_i|e_i\rangle \quad \hat{B}|e_i\rangle = \mu_i|e_i\rangle \quad \Leftrightarrow \quad \langle e_i|\left[\hat{A}, \hat{B}\right]|e_j\rangle = \left(\lambda_i\mu_i - \lambda_j\mu_j\right)\langle e_i|e_j\rangle = 0$$

Canonical commutator

$$[\hat{p}, \hat{x}] = -i\hbar$$

QM — Uncertanty Principle

We define the **uncertainty** of an operator $\hat{A}$ in a state $|\psi\rangle$ as its **standard deviation**:

$$\Delta^2 \hat{A}_\psi = \langle \hat{A}^2 \rangle_\psi - \left(\langle \hat{A} \rangle_\psi \right)^2 = \langle \left(\hat{A} - \langle \hat{A} \rangle_\psi \right)^2 \rangle_\psi$$

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The uncertanty of a pair of operators must satisfy the **uncertainty principle**:

$$\frac{1}{4} \, |\langle \left[\hat{A}, \hat{B} \right] \rangle_\psi|^2 \leq |\langle \hat{A}\hat{B} \rangle_\psi|^2 \leq \langle \hat{A}^2 \rangle_\psi \langle \hat{B}^2 \rangle_\psi$$

Heisemberg's Principle

$$\Delta^2\hat{x}\Delta^2\hat{p} \geq \frac{\hbar^2}{4}$$

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Spectrum

$$\hat{\boldsymbol{L}}^2 |lm\rangle = \hbar^2 l(l+1) |lm\rangle \quad l \in \mathbb{N}$$

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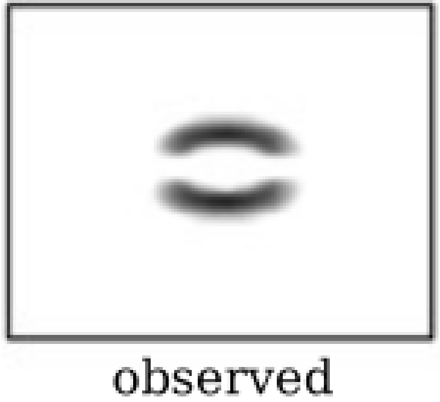
Degeneracy $\quad d_l = 2l + 1$

QM — Theoretical Review



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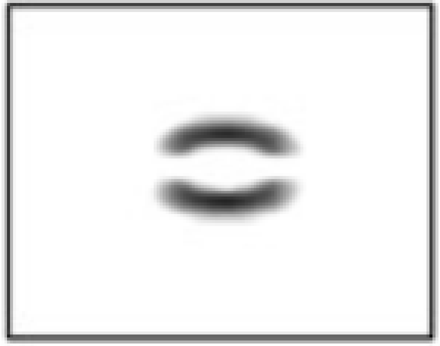

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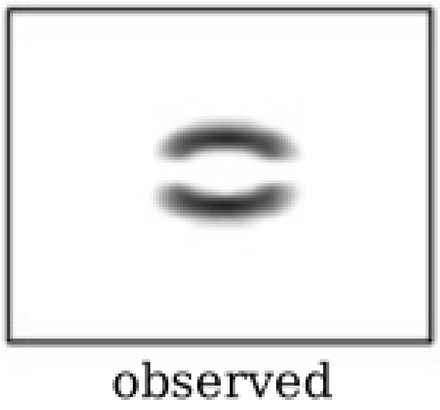

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But $\hat{L}_z$ has $d_l = 2l + 1$ possible eigenvalues, which is **odd**... In the case $l = 0$ we should see just one undeflected spot.

Why two deflected spots?

Spin

QM — Intrinsic Angular Momentum

Commutation rules

$$\left[\hat{S}_i, \hat{S}_j\right] = i\hbar\varepsilon_{ijk}\hat{S}_k \qquad \left[\hat{S}_i, \hat{\boldsymbol{S}}^2\right] = 0$$

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Energy Spectrum

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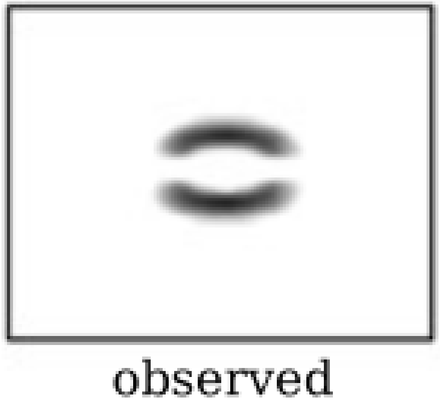
Degeneracy $\quad d = d_l d_s = (2l+1)(2s+1)$

QM — Experimental Outcome



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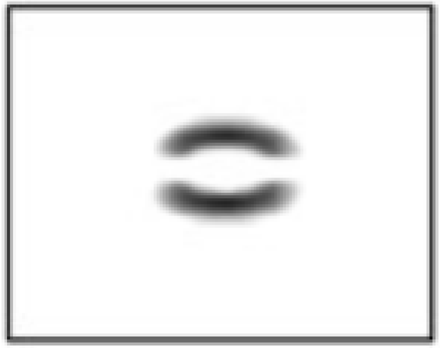

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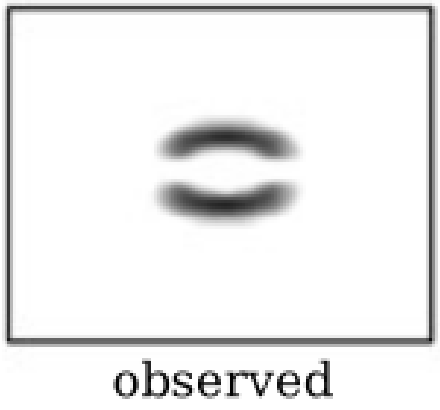

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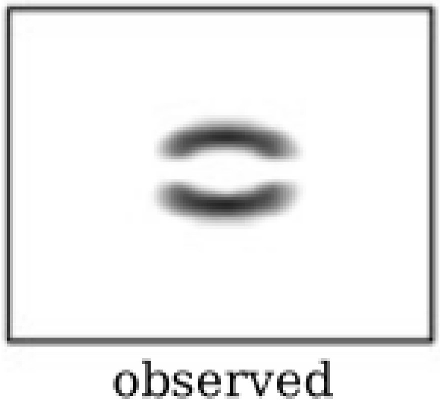
the SG apparatus is **measuring the spin!**

QM — Experimental Outcome



The degeneracy is $d = d_s = 2s + 1 = 2$

$$s = \frac{1}{2} \quad \Rightarrow \quad m_s = \pm\frac{1}{2}$$

the SG apparatus is **measuring the spin!**

By measuring the spacing on the screen, from the energy spacing

$$\Delta E_{0,\pm\frac{1}{2}} = g_s \mu_B B_z$$

we also conclude that $g_s = 2$

Conclusion

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Conclusion

- SG apparatus measures the component of **magnetic momentum** in the field-gradient direction
- To describe the experimental results it was introduced the **spin**, a new degree of freedom with no classical analoguos.
- Spin can assume both **integer** and **half-integer** eigenvalues.
- Particles are classified by their spin as **bosons** or **fermions**.
- Also **nuclei** and **molecules** have a magnetic momentum.

References

- *Introduction to the Physics of Matter*, Nicola Manini
- *Fisica quantistica*, Stefano Forte, Luca Rottoli

Thank you for the attention!

Total Angular Momentum

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

Definition

$$\hat{\boldsymbol{J}} = \hat{\boldsymbol{L}} + \hat{\boldsymbol{S}}$$

$$\hat{\boldsymbol{J}}^2 = \hat{\boldsymbol{L}}^2 + \hat{\boldsymbol{S}}^2 + 2\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{L}}$$

Commutation rules

$$\left[\hat{L}_i, \hat{S}_j\right] = 0 \qquad \left[\hat{J}_i, \hat{\boldsymbol{J}}^2\right] = 0$$

Spectrum

$$\hat{\boldsymbol{J}}^2|jm_j\rangle = \hbar^2 j(j+1)|jm_j\rangle \quad |l-s| \leq j \leq l+s$$

$$\hat{J}_z|jm_j\rangle = \hbar m_j|jm_j\rangle \quad -j \leq m_j \leq j$$

Degeneracy

$$d_j = 2j + = (2l+1)(2s+1)$$

Coupling of Angular Momenta

Consider the magnetic field experienced by an e^- orbiting around a nucleus with Z p^+

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{Z q_e}{4\pi\epsilon_0 c^2 m_e} \frac{\hat{\boldsymbol{L}}}{|\boldsymbol{r}|^3}$$

Spin-Orbit coupling $\quad \hat{H}_{\text{s-o}} = -\frac{1}{2}\hat{\boldsymbol{\mu}}_s \cdot \boldsymbol{B}$

$$\begin{aligned}\hat{H}_{\text{s-o}} &\propto \frac{1}{r^3}\hat{\boldsymbol{S}} \cdot \hat{\boldsymbol{L}} \\ &\propto \frac{1}{r^3}\left(\hat{\boldsymbol{J}}^2 - \hat{\boldsymbol{L}}^2 - \hat{\boldsymbol{S}}^2\right)\end{aligned}$$

We can represent the e^- state using the basis

$$\left\{\hat{\boldsymbol{L}}^2, \hat{L}_z, \hat{\boldsymbol{S}}^2, \hat{S}_z\right\}$$

Another option is the **coupled** basis

$$\left\{\hat{\boldsymbol{J}}^2, \hat{J}_z, \hat{\boldsymbol{L}}^2, \hat{\boldsymbol{S}}^2\right\}$$

since

$$\left[\hat{J}_z, \hat{\boldsymbol{L}}^2\right] = \left[\hat{J}_z, \hat{\boldsymbol{S}}^2\right] = 0$$

$$\left[\hat{\boldsymbol{J}}^2, \hat{\boldsymbol{L}}^2\right] = \left[\hat{\boldsymbol{J}}^2, \hat{\boldsymbol{S}}^2\right] = 0$$

Energy Spectra in Magnetic Field

Paschen-Back Effect $\mu_B B \gg \xi$

$$\xi_{m_l m_s} = \mu_B B_z (m_l + 2m_s)$$

- Spin-orbit coupling negligible
- Good quantum numbers: m_l, m_s
- Linear splitting

Zeeman Effect $\mu_B B \ll \xi$

$$\xi_{m_j} \approxeq g_j m_j \mu_B B_z$$

- Spin-orbit coupling dominant
- L and S coupled to J
- Good quantum numbers: j, m_j